

SAIFUL ISLAM

Computer Science & Engineering Undergraduate • Mobile & Web Developer

Sylhet, Bangladesh • +880 1707-224860 • saiful1616.islam@gmail.com • github.com/saiful16164 • linkedin.com/in/saiful1616

EDUCATION

Sylhet Engineering College — B.Sc. in Computer Science & Engineering

2022 – Present

Sylhet, Bangladesh

- **Coursework:** Data Structures & Algorithms, OOP, DBMS, Software Engineering, Machine Learning, Operating Systems, Computer Networks.

BAF Shaheen College Shamshernagar — Higher Secondary Certificate (HSC), Science

2020 – 2022

Moulvibazar, Bangladesh

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Dart, Kotlin, TypeScript, JavaScript, SQL
Frameworks & Libraries: Flutter, Android SDK, React.js, Next.js, PyTorch, Node.js, Tailwind CSS
Databases & Cloud: PostgreSQL, Supabase, Firebase (Firestore, Auth, FCM), SQLite
Developer Tools: Git, GitHub, VS Code, Android Studio, Linux, RESTful APIs, Postman

RESEARCH PUBLICATIONS

Classification of Guava and Mango Leaf Diseases Using YOLOv11 and Stable Diffusion-Based Augmentation

Dec 2025

2025 28th International Conference on Computer and Information Technology (ICCIT) | DOI: [10.1109/ICCIT68739.2025.11491324](https://doi.org/10.1109/ICCIT68739.2025.11491324)

- Formulated a deep learning pipeline pairing YOLOv11 with generative Stable Diffusion data augmentation to eliminate agricultural dataset scarcity.
- Achieved superior detection accuracy and recall across complex plant pathologies under uncontrolled field conditions.

Bangla Optical Character Recognition using Vision Transformers and CNN Ensembles

Dec 2025

2025 IEEE WIECON-ECE Conference | DOI: [10.1109/WIECON-ECE69386.2025.11526313](https://doi.org/10.1109/WIECON-ECE69386.2025.11526313)

- Designed a hybrid Vision Transformer (ViT) and CNN architecture for high-accuracy recognition of complex cursive and printed Bangla scripts.
- Benchmarked Character Error Rates against existing baselines, demonstrating elevated robustness to noisy and low-resolution inputs.

KEY PROJECTS

Smart Ledger — Personal Finance & Expense Analytics App

Flutter • Dart • Supabase • PostgreSQL

- Engineered a cross-platform mobile expense tracker featuring categorized transaction logging, budget alerts, and real-time cloud sync.
- Architected offline-first local persistence with Supabase background syncing, maintaining sub-50ms UI response times without network.
- Built interactive statistical visualizers (breakdown charts, monthly trends, multi-currency conversion) for actionable financial insights.

SEC Routine App — Automated College Schedule & Timetable Manager

Flutter • Firebase • Kotlin • Android

- Developed an academic scheduling application adopted by Sylhet Engineering College students to automate lecture tracking and timetable lookups.
- Integrated Firebase Cloud Messaging (FCM) to trigger background push notifications prior to scheduled lecture and lab sessions.
- Implemented personalized schedule filtering by department, semester, and section with offline timetable caching.

Exam Rush Hour — Real-Time Academic Assessment Platform

Next.js • TypeScript • React • PostgreSQL

- Architected an interactive examination platform featuring timed challenge sessions, peer competition, and live score leaderboards.
- Constructed high-performance server-rendered UI components with Next.js App Router and secure user session management.

HONORS & ACTIVITIES

SUST Hackathon 2026 — Shahjalal University of Science & Technology (SUST EEE Carnival)

2026

- Collaborated in an intensive sprint environment to architect, build, and pitch technical solutions under competitive deadlines.

Leading University Junior IUPC 2024 — Inter-University Programming Contest

2024

- Represented Sylhet Engineering College solving complex algorithmic, dynamic programming, and data structure problem sets.

Paper Presenter — IEEE ICCIT 2025 & IEEE WIECON-ECE 2025

Dec 2025

- Delivered oral presentations defending computer vision research at international IEEE conferences in Cox's Bazar, Bangladesh.